

Math5485

Introduction to Numerical Methods I

Jack Thomas

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Table of contents

Welcome!	3
I Lectures	4
1 Introduction	5
1.1 Subheading	5
II Appendices	6
III Problem Classes	7
IV Assignments	8
V Exams	9
Appendices	10
VI Admin	10
A Course Syllabus	11
Resources	12

Welcome!

These are the lecture notes for Math 5485: Introduction to Numerical Methods I, Fall 2026.

Use the links on the side or navigate by chapter:

- [Chapter 1: An Introduction to Numerical Analysis](#)
- [Chapter 2: Solving Nonlinear Equations in 1d](#)
- [Chapter 3: Polynomial Interpolation](#)
- [Chapter 4: Numerical Integration](#)
- [Chapter 5: Linear systems of equations](#)

Useful links:

[Syllabus](#) [GitHub](#) [Fall 2025 Course](#) [Ask a Question](#) [Suggest Correction](#) [Math 5486](#)

References

- [Driscoll and Braun \(2017\)](#)
- [Burden et al. \(2016\)](#)
- [Süli and Mayers \(2003\)](#)
- [Trefethen \(2019\)](#)

Part I

Lectures

1 Introduction

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(knuth84?)

1.1 Subheading

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Part II

Appendices

Part III

Problem Classes

Part IV

Assignments

Part V

Exams

Part VI

Admin

A Course Syllabus

Resources

- Burden, Richard L., J. Douglas Faires, and Annette M. Burden. 2016. *Numerical Analysis*. 10th ed. Cengage Learning. <https://doi.org/10.13140/2.1.4830.2406>.
- Driscoll, Tobin A., and Richard J. Braun. 2017. *Fundamentals of Numerical Computation*. Society for Industrial; Applied Mathematics. <https://doi.org/10.1137/1.9781611975086>.
- Süli, Endre, and David F. Mayers. 2003. *An Introduction to Numerical Analysis*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511801181>.
- Trefethen, Lloyd N. 2019. *Approximation Theory and Approximation Practice, Extended Edition*. Society for Industrial; Applied Mathematics. <https://doi.org/10.1137/1.9781611975949>.